Capital University of Science & Technology

Term Project Proposal

Department of Electrical and Computer Engineering

Project Title	**Password-Based Digital Access Control System**	
Course Title	**Digital Logic Design (DLD)**	
S#	**Student Name**	**Registration Number**
1	Aetisaam ul Hassan	BCY253051
2	Aman Sajid	BCY253084
3	Maier Masood	BCY253033

Project Idea: Password-Based Digital Access Control System

The project is a **Dual-Mode Password-Based Digital Access Control System** designed using Digital Logic Design concepts. It allows access only when the correct password is entered through input switches on a breadboard. The system has two separate passwords: one for the **admin** and one for the **normal user**. Logic ICs are used to compare the entered password with the preset password. If the password is correct, a green LED turns ON, and if the password is wrong, a red LED turns ON along with a **buzzer alarm**.

Project Objective:

The main objective of this project is to design and implement a **Dual-Mode Password-Based Digital Access Control System** using Digital Logic Design concepts.

The project aims to:

- Design a password verification system using **logic gates and ICs**.
- Implement the circuit practically on a **breadboard**.
- Provide two separate password modes: one for **admin** and one for **normal user**.
- Compare the entered password with the preset password using combinational logic.

- Indicate correct password entry using a **green LED**.
- Indicate wrong password entry using a **red LED**.
- Activate a **buzzer alarm** whenever an incorrect password is entered.
- Demonstrate the practical use of DLD concepts such as **truth tables, Boolean expressions, logic gates, and IC-based circuit implementation**.

Terminologies Used:

- **Digital Logic Design:** Designing circuits using binary values 0 and 1.
- **Binary Input:** Password entered using switches as 0 or 1**.**
- **Logic Gates:** Basic gates like AND, OR, NOT, XOR/XNOR used for decision-making.
- **ICs:** Chips that contain logic gates for circuit implementation.
- **Breadboard:** A board used to build circuits without soldering.
- **Password Verification:** Checking whether the entered password is correct or wrong.
- **Admin Password:** Separate password for admin access.
- **User Password:** Separate password for normal user access.
- **Mode Selection:** Choosing between admin mode and user mode.
- **Comparator Logic:** Logic used to compare entered password with preset password.
- **Truth Table:** Table showing input combinations and outputs.
- **Boolean Expression:** Logical equation used to design the circuit.
- **LED Indicator:** Green LED shows correct password, red LED shows wrong password.
- **Buzzer:** Rings when a wrong password is entered.
- **Combinational Circuit:** Circuit whose output depends only on current inputs.

METHODOLOGY:

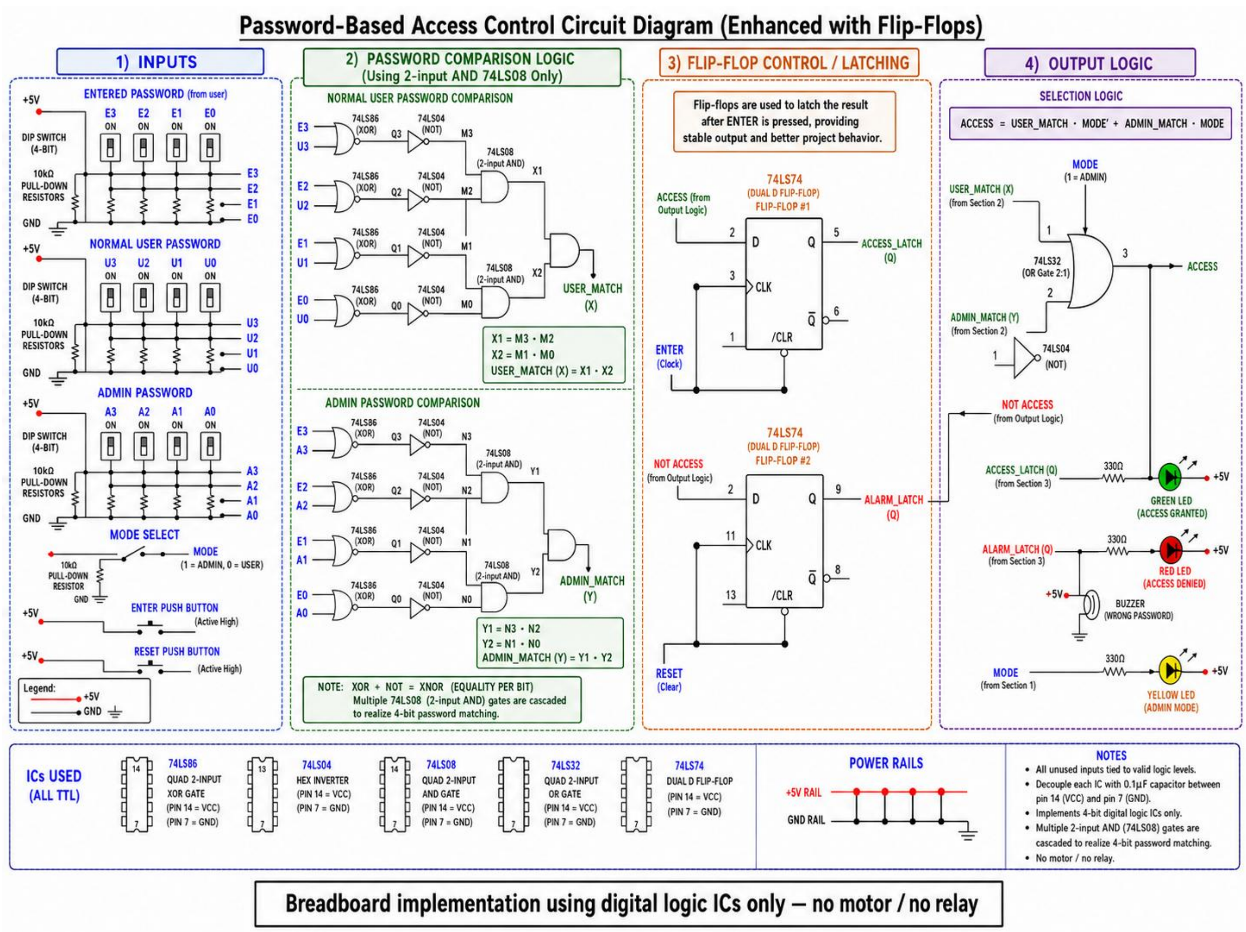
The project will be designed as a **breadboard-based digital password verification system** using basic logic ICs. First, separate binary passwords will be assigned for the **admin** and **normal user** modes. Input switches will be used to enter the password, and a mode selection switch will select whether the system checks the admin password or the user password.

After that, a truth table will be created for all possible input combinations. Boolean expressions will be derived from the truth table and simplified using DLD concepts. These simplified expressions will then be implemented using logic gates such as AND, OR, NOT, XOR/XNOR gates through ICs.

The entered password will be compared with the selected preset password using comparator logic. If the password matches, the **green LED** will turn ON to show successful access. If the password is incorrect, the **red LED** will turn ON and the **buzzer** will ring to indicate wrong password entry.

Finally, the complete circuit will be implemented and tested on a breadboard using a 5V power supply, logic ICs, resistors, LEDs, switches, and buzzer. The output will be verified by testing both admin and user passwords.

CIRCUIT DIAGRAM:



Application of the Project:

- **Digital Door Lock:** Allows entry only with the correct password.
- **Locker Security:** Protects lockers from unauthorized access.
- **Lab/Office Access:** Can be used for restricted areas.
- **Alarm System:** Buzzer rings when a wrong password is entered.
- **Dual-Mode Security:** Provides separate access for admin and normal user.
- **DLD Learning:** Demonstrates logic gates, ICs, truth tables, and Boolean expressions practically.

ICs and Components used:

Table 1: ICs Used

Sr. No.	IC Number	IC Name	Purpose
1	74LS86	Quad 2-Input XOR Gate	Compares each entered password bit with the stored password bit.
2	74LS04	Hex NOT Gate / Inverter	Converts XOR outputs into XNOR/equality logic and provides required NOT signals.
3	74LS08	Quad 2-Input AND Gate	Cascaded 2-input AND gates combine equality outputs; no 4-input AND gate is used.
4	74LS32	Quad 2-Input OR Gate	Combines user-match and admin-match paths in the final access selection logic.
5	74LS74	Dual D Flip-Flop	Latches ACCESS_GRANTED and ALARM outputs after ENTER is pressed until RESET is applied.

Table 2: Other Components Used

Sr. No.	Component	Quantity	Purpose
1	Breadboard	1 large	For complete circuit implementation.
2	5V DC Power Supply	1	Provides regulated power to TTL ICs and other components.
3	4-bit DIP Switch	3	Used for entered password, normal user password, and admin password.
4	Mode Select Switch	1	Selects normal user mode or admin mode.
5	Green LED	1	Indicates access granted.
6	Red LED	1	Indicates access denied / wrong password.
7	Yellow LED	1	Indicates admin mode.
8	Buzzer	1	Rings when wrong password is entered.
9	330 ohm Resistors	3	Current-limiting resistors for LEDs.

10	10k ohm Resistors	As required	Pull-down resistors for switches and push buttons.
11	0.1 uF Capacitors	One per IC	Decoupling capacitors for stable IC operation.
12	Jumper Wires	As required	For making breadboard connections.

Table 3: Important Design Notes

Point	Description
Power convention	For each 14-pin TTL IC, pin 14 is connected to +5V and pin 7 is connected to GND.
No 4-input gates	The circuit uses cascaded 2-input 74LS08 AND gates to perform 4-bit password matching.
No motor / relay	The project output is shown using LEDs and a buzzer only; no motor or relay is included.
Flip-flop	74LS74 flip-flops latch the result after ENTER is pressed, keeping the output stable until RESET is pressed.

Student 1 Sig. ____________________ Instructor Sig: ____________________

Student 2 Sig. ____________________

Student 3 Sig. ____________________